

Ben Nguyen

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Research Interests

Computational immunology; functional genomics and CRISPR screen analysis; T-cell biology and cell therapy (CAR-T); single-cell / Perturb-seq analysis; and reproducible, provenance-tracked data pipelines.

Education

Archbishop Mitty High School

High School Diploma — Class of 2027 (Grade 12)

San Jose, CA

Expected 2027

West Valley College — Dual Enrollment

College coursework completed during high school

Saratoga, CA

2025 – 2027

Research Experience

Reina Lab, La Jolla Institute for Immunology

Volunteer Research Contributor

Aug 2026 – Present

Hybrid

- Re-analyze published CRISPR screens, including single-cell Perturb-seq, with MAGeCK; curate results for FITdb, the lab's immunogenomics database.
- Build reproducible notebooks with quality checks and data provenance; contributed a single-cell Perturb-seq analysis deliverable.

UC Berkeley Bioengineering Honor Society (BioEHSC 2026)

Project Lead & Biology Specialist

Jan – Apr 2026

Hybrid

- Led a five-person team that proposed a dual-layer quality-control system for CAR T-cell manufacturing using Raman spectroscopy, spectral flow cytometry, and SCENITH metabolic profiling.
- Translated T-cell biology into engineering requirements; wrote a 23-page architecture guide and compiled a 13-paper citation library.

ThinkNeuro, LLC

Research Intern

Feb – Apr 2026

Remote

- Completed a 10-week research program; analyzed neural-circuit literature in R, earned a Certificate in R and Bibliometrics, and presented at the ThinkNeuro Symposium.

Publications & Manuscripts

Nguyen, B. (first author). “AI-Driven Dual-Layer Quality-Control System for CAR T-Cell Manufacturing.” Manuscript in preparation. Describes a proposed quality-control architecture; it has not been experimentally or clinically validated.

Presentations

- “Dual-Layer QC System for CAR T-Cell Manufacturing.” UC Berkeley BioEHSC 2026.
- “Bibliometric Meta-Analysis of a Neural Circuit.” ThinkNeuro Symposium, 2026.

Honors & Awards

- **1st Place, Biological Engineering** — UC Berkeley BioEHSC 2026 (78 teams, including international teams).
- **1st Place, CED Hacks 2026** — BinIt team; \$100 first-place prize.
- **Monarch Award, Wrestling** — Archbishop Mitty High School, sophomore year (2024–25).

Certifications

- Certificate in R and Bibliometrics — ThinkNeuro, 2026.

Technical Skills

Programming & Data	Python, PyTorch, R, data curation, bibliometrics, Node.js, Express
Bioinformatics	CRISPR screen analysis (MAGeCK), single-cell Perturb-seq, reproducible pipelines
Methods (system design)	Raman spectroscopy, spectral flow cytometry, SCENITH metabolic profiling, cross-modal ML
Domain	Immunology, T-cell biology, cell therapy / CAR-T, immunogenomics

Selected Projects

FITdb — Immunogenomics Research Database **Reina Lab**

- Curate published CRISPR knockout, CRISPRi/a, and single-cell Perturb-seq screens; re-analyze results with MAGeCK and build reproducible notebooks with quality checks and recorded data provenance.

BinIt **CED Hacks, 2026**

- Built a photo-based waste-sorting app with two teammates. It assesses an item's material and condition, including grease on a paper plate, checks local disposal guidance, and returns the municipal source; the demo used San Jose rules.

Cooperative AI Engine for Bughouse Chess **Python, PyTorch**

- Building with a classmate under PhD mentorship; a cooperative engine that plays one board as a human teammate in a two-board chess variant under partial information and time pressure.

Leadership & Service

Varsity Wrestling Captain. Encourage teammates to review matches honestly.

Martial Arts Club Founder & President. Started the club junior year to build a student community around martial arts.

Vietnamese Student Association Officer. Helped grow the club into one of the school's largest.

Skill Toys Club Co-founder. Started the club with a friend for students who share the hobby.

Tamayo's Judo Youth Coach. Teach judo to younger students.

Second Harvest of Silicon Valley Food Distributor, Apr 2023–Present. Distribute fresh groceries, assist families with selections, manage inventory and restocking.

Catholic Charities USA Food Distributor, Apr 2023–Present. Support disaster and humanitarian relief.